

John D. Duncan, III

Phone | (240) 688-7187
Location | Washington, DC
Status | US Citizen

Email | cv@johnduncan.io
Homepage | <https://johnduncan.io>
Github | <https://github.com/johndduncanii>
Linkedin | <https://linkedin.com/in/johndduncan>

PROFESSIONAL SUMMARY

Quorum Analytics

Washington, DC

Wrote and launched over a dozen flagship products for Quorum's best-in-class public affairs platform, helping drive growth from \$9M to \$64M+ ARR by consistently solving the team's most complex technical challenges while delivering projects on time and under budget. During my tenure, I authored 5k+ commits, 1k+ code reviews, 700+ PRs, conducted 100+ interviews, and mentored dozens of engineers. I recently launched PAC Hybrid, expanding Quorum into the \$15M+ PAC market, where I:

- Discovered critical discrepancies in Federal Election Commission (FEC)-designated national party committee C-number ids officially listed across 50+ years of legal advisory opinions, FEC website data, and official fec-cms GitHub repository.
- Developed end-to-end search, filtering, and data visualization features for four datasets, including backend sql queries, API endpoints, and frontend UI functionality.
- Designed a database synchronization infrastructure to instantly sync billions of rows between Quorum and legacy CisionPAC databases.
- Reduced database query latency from 60 seconds to 20 milliseconds by optimizing Postgres partition and schema constraints, improving transaction dataset query performance by over 3000x and permanently fixing a 10+ year old bug.
- Coached engineers to optimize a database migration by replacing sequential queries with a single constant-time operation, reducing execution time from two weeks to under five seconds and improving backfill performance by over 240,000x.

My full CV with a detailed project history is available on my website: <https://johnduncan.io/cv.pdf>.

Massachusetts Institute of Technology: Lincoln Laboratory (MIT: LL)

Lexington, MA

During my internship at MIT Lincoln Laboratory, I wrote Local Evacuation Alert Verification (LEAV), an Android application sponsored by the Department of Homeland Security (DHS) Science and Technology (S&T) Directorate which routes hurricane evacuees to shelters, displays real-time evacuation zone status through an interactive map, and facilitates communication between State Emergency Managers (EMs) and evacuees. I also wrote an interactive module for HURREVAC-eXtended (HVX), a project sponsored by the Federal Emergency Management Agency (FEMA) National Hurricane Program (NHP) Technology Modernization initiative, enabling EMs to update and monitor the status of evacuation zones via a web interface.

LEADERSHIP

The Cairo

Washington, DC

Outside of work, I recently co-led a board transformation for The Cairo (1894), a historic building in DC with a \$1M+ annual budget where I currently serve as Vice President and Secretary of the Board of Directors. As VP, I co-developed a strategic financial management plan by auditing finances, eliminating wasteful spending, drafting board policies, restructuring staff, enhancing vendor accountability, and developing a proactive capital improvement strategy—all while keeping fee increases below 5%.

Last year, I partnered with a horticulturist from Harvard's Dumbarton Oaks Gardens and a landscape architect from Oehme, van Sweden & Associates to design and install a perennial pollinator garden, incorporating heirloom plants that are concomitant with period-appropriate Victorian varieties.

The Cairo, which is the tallest privately owned building in the city, is listed on both the DC Inventory of Historic Sites (1990) and National Register of Historic Places (1994).

Eisenhower Institute

Gettysburg, PA

During my undergraduate studies, I served as an Eisenhower Institute Undergraduate Fellow (EIUF), a selective program designed to cultivate political leadership skills and empower students to tackle society's most challenging public policy issues.

PERSONAL PROJECTS

In my free time, I enjoy challenging myself by working on idiosyncratic projects across diverse domains. I recently built a Golang query builder compatible with Django ORM syntax, uncovered a bug in the Google Chromium URI parser, discovered a novel method to losslessly convert legacy Apple Macintosh data fork fonts to macOS resource forks, ported James Ashton's X-Face decoder from C to Golang, and created a Mozilla XUL/JavaScript add-on to display 'face' images from email and newsgroup message headers. I also have a passion for design and previously won a Gentoo Linux interface design competition.